

GeoQuery

Natural Language Understanding of RGB Imagery

MID PREP | PROBLEM STATEMENT
Inspired by ISRO SAC — Inter IIT Tech Meet 14.0

1. Background & Motivation

We live in a world saturated with visual information. Cameras mounted on drones, aircraft, and everyday devices capture millions of overhead and street-level images every hour. Yet the ability to extract structured, queryable knowledge from these images remains locked behind expensive annotation pipelines and domain-specific software tools.

Recent advances in Vision-Language Models (VLMs) have demonstrated impressive zero-shot capabilities: given an image, these models can describe scenes, locate objects, and answer questions — all through natural language. However, bridging the gap between raw model output and a usable, reliable application requires thoughtful system design, grounding techniques, and careful evaluation.

This problem statement challenges you to build exactly that bridge: a deployable, interactive application that lets any user upload a standard RGB photograph and instantly obtain structured visual intelligence — captions, bounding boxes, and question answering — through a conversational interface.

2. Problem Statement

Design and build a functional, interactive web application that enables non-expert users to interrogate RGB images through natural language. The system must support the following three core capabilities:

2.1 Image Captioning

When an image is uploaded, the system must automatically generate a concise, accurate natural language caption that summarises the key content of the image — objects present, their spatial arrangement, and any notable scene attributes. The caption should be factual, specific, and suitable for a non-expert audience.

2.2 Object Detection with Bounding Boxes

The system must detect and localise objects belonging to a predefined set of six classes (see Section 4) within the uploaded image. Detected objects must be visually highlighted using axis-aligned bounding boxes overlaid on the image, with class labels and confidence scores displayed.

Detection must be triggered either automatically upon upload or in response to a natural language query (e.g., "Mark all vehicles in the image").

2.3 Visual Question Answering (VQA)

Users must be able to ask free-form natural language questions about the image or about specific detected objects. The system should correctly handle three categories of questions:

- Binary questions (yes/no): e.g., "Is there a road visible in this image?"
- Numeric questions: e.g., "How many vehicles are parked in the lot?"
- Attribute questions: e.g., "What colour is the rooftop of the largest building?"

All three capabilities must be accessible through a unified chat interface. The user should not need to switch between modes — the system should infer intent from the query.

3. Technical Scope & Constraints

3.1 Input Specification

- Image format: Standard RGB images in .jpg or .png format only.
- Maximum image resolution: 1024 x 1024 pixels. Images exceeding this must be gracefully resized or rejected with a clear error message.
- No multi-spectral, SAR, or infrared imagery is required. Pure three-channel (R, G, B) images are the only input modality.
- Images may depict urban scenes, natural landscapes, mixed environments, or any combination of the six target classes.

3.2 Allowed Resources

- Pre-trained open-source VLMs and object detection models are permitted (e.g., CLIP, LLaVA, Grounding DINO, YOLO variants, SAM). Provide full credits.

- Fine-tuning on publicly available datasets is encouraged and will be rewarded in the innovation component.
- Direct use of paid commercial APIs (OpenAI Vision, Google Vision AI, AWS Rekognition, etc.) as the primary inference backbone is NOT permitted.
- Open-source inference via locally hosted model weights is permitted.

3.3 Deliverables

1. A working web application (hosted or locally runnable) with a chat-based interface.
2. Complete source code (frontend + backend) submitted as a compressed archive or repository link.
3. A brief technical report (maximum 6 pages, double-column format) covering system design, model choices, and results.
4. A 5-minute live demonstration during the final evaluation session.

4. Target Object Classes

Detection, grounding, and attribute queries must be constrained to the following six classes. Teams are not required to detect objects outside this set; the system should gracefully indicate when a queried object does not belong to the supported classes.

Class Label	Description	Example Query
building	Residential or commercial structure identifiable from overhead view	"How many buildings are visible?"
road	Paved surface used for vehicular movement	"Is there a road in the lower half?"
vehicle	Car, bus, truck, or other motorised transport visible from above	"Count the vehicles in the parking lot."
vegetation	Trees, grass patches, crop fields, or forested area	"What fraction of the image is vegetation?"
water body	River, lake, pond, or other standing/flowing water	"Identify and mark all water bodies."
open ground	Bare land, sandy patches, or unpaved open area	"Describe the open ground region."

Note: A single image may contain zero, one, or many instances of each class. Overlapping objects (e.g., a vehicle partially obscured by vegetation) should be handled as best as the model permits, but graceful degradation is acceptable.

5. Interface Requirements

The chat interface must satisfy the following minimum requirements:

- Image upload: A clearly visible upload button or drag-and-drop zone.
- Automatic caption: Displayed immediately after upload, before any user query.
- Bounding box overlay: Rendered on the image in the chat window, not as a separate download.
- Object descriptions: Each detected object should be accompanied by a short textual description (class, confidence, and one key attribute such as colour or size estimate).
- QnA responses: Text responses to user queries displayed in a chat bubble format.
- Session history: The conversation history (queries + responses) should be visible within the session.
- Error handling: Clear, user-friendly messages for unsupported file types, resolution overflows, or out-of-scope queries.

Bonus (not required, but rewarded under innovation):

- Multi-turn context: The system retains context from previous turns (e.g., "How many of those vehicles are trucks?")
- Object-level queries: Clicking on a detected bounding box triggers a follow-up description of that specific object.
- Exportable report: Option to download the annotated image and Q&A transcript as a PDF.

This is a simplified mock problem statement prepared by IIT Ropar for internal preparatory purposes. It is inspired by, but not identical to, the original ISRO SAC problem statement from Inter IIT Tech Meet 14.0. [View Original Problem Statement](#)

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